

Analysis of Aggregation Algorithms in Federated Learning for Non-Identically Distributed Data

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Abstract—The paper covers issues of studying algorithms for aggregating local results for non-identically distributed data within the framework of federated machine learning. Federated learning replaces centralized data processing, reducing the load on the infrastructure and providing increased information security. Data is processed directly on client devices, eliminating the need for their transmission. Due to the fact that federated learning is characterized by various types of heterogeneity in data, the problem of the quality of processing non-identically distributed and dependent data arises. The purpose of this work is to analyze existing and promising aggregation algorithms within the federated learning and to study the challenges caused by the heterogeneity of data distribution between participants in the learning process. The novelty of the study is presented by a comprehensive consideration of the potential of aggregation methods in relation to the cases with unbalanced data sets. The main challenges identified relate to the lack of a unified classification and criteria for assessing non-identically distributed data, universal metrics for data heterogeneity, and consideration of side effects when creating shifts in data. Generally, it necessitates the unification of approaches to analyzing and accounting for data heterogeneity to improve the quality of trained models. Based on the results of this work, subsequent studies are expected to be aimed at developing adaptive aggregation strategies, including dynamic changes in hyperparameters and the introduction of client clustering mechanisms to improve the efficiency of federated learning in the presence of non-identical data.

Keywords—federated learning, aggregation, non-identically distributed data, analysis, clustering

I. INTRODUCTION

The concept of federated learning (FL) emerged as an alternative to centralized learning, which involves collecting data from all sources of a distributed system into a single repository, processing it, and then analyzing it. Federated learning is designed to solve such problems as reducing the volume of transmitted and stored data, decreasing the performance requirements of the central server, and increasing data security. In addition, federated learning is aimed to ensure the privacy and confidentiality of user data, without the need to transfer it to any other network hosts and devices. Also, the advantages of federated learning include increased flexibility and resilience of such machine

learning mechanisms to various communication failures, including those caused by attackers.

In general, federated learning includes the following stages: initialization of the global model, selection of clients for a separate round, distribution of the global model among the selected clients, local training of models, aggregation of local models, and obtaining an updated general model.

The basic federated learning algorithm FedAvg [1] includes such steps as initialization of a global model, client selection for the training round, model distribution, local training, model aggregation and model averaging. If we have a subsets S_t of K devices in a round and each device $K \in S_t$ sends the model $\mathbf{w}_k^{t+1}$ after a local training, the server aggregates the new global model $\mathbf{w}$ as in Eq. (1):

$$\mathbf{w}^{t+1} = \frac{1}{K} \sum_{k \in S_t} \mathbf{w}_k^{t+1} \quad (1)$$

Note, that model averaging is a suitable solution if all models received from clients are equal, i.e. they have similar architecture and determined by the same set of weights.

Specifically, one should notice, federated learning can be characterized by various types of heterogeneity. The heterogeneity could occur on the data distribution level, device computational resources and device availability during training. Some researchers outline also heterogeneity on the task and model level [2]. However, one of the most complicated problem is the heterogeneity at the data level. It consists it that client data may differ in the size of the training sample, the distribution of labels, and the distribution of attribute values. Therefore, when using different data sets belonging to different clients, the problem of handling non-identically distributed and dependent data becomes especially important.

Label distribution skew refers to the variation in the distribution $P(Y)$ of labels Y across different clients. In typical FL experimental setups, datasets are sorted by labels and assigned to clients in a round-robin fashion. This means that client i may have data labeled as “class1” and “class2”, while client j has data labeled as “class3” and “class4”, with no label overlap between clients. Feature distribution skew refers to the variation in the distribution of features, denoted as $P(x)$ or input data, across different

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clients in FL. This variation can lead to discrepancies in the representation of data for the same class across different clients.

This paper reviews and analyzes local model aggregation algorithms in the case of federated learning of non-identically distributed data, features, as well as studies advantages and disadvantages of such algorithms. The main contribution of this work is presented by an identification and analysis of the revealed challenges and problems associated with non-identical data distribution among clients within the framework of federated learning. The novelty of the work includes a comprehensive consideration of the capabilities of aggregation algorithms for the case of non-identically distributed data, which is insufficiently reflected in existing scientific works.

The rest of the paper is organized as follows. Section 2 presents an analysis of the main existing aggregation algorithms within federated learning, which have been proposed to date and tested for solving a number of applied problems of distributed data analysis. Section 3 discloses the main conclusions made on the basis of the analysis and is focused on challenges associated with the characterization, use and improvement of the quality of data analysis in aggregation algorithms.

II. ANALYSIS OF AGGREGATION ALGORITHMS

Distributed systems often have a complex architecture and consist of a large number of heterogeneous devices. Some systems also assume geographic distribution. The application of the FedAvg algorithm to non-identically distributed data has a negative impact on the accuracy of models [2], therefore, in systems characterized by data heterogeneity, other algorithms must be used.

There are algorithms for aggregating local models that take into account the features of non-identically distributed data. Most of them are based on FedAvg and are its improvements [3].

The difference between FedProx [4] and FedAvg consists in the way how to train local models, namely in adding proximal term to the local subproblem to effectively limit the impact of variable local updates. In particular, instead of just minimizing the local function $F_k(\cdot)$, device k uses its local solver of choice to approximately minimize the following objective h_k "(2)":

$$\min_w h_k(\omega; \omega^i) = F_k(\omega) + \frac{\mu}{2} \|\omega - \omega^i\|^2 \quad (2)$$

The coefficient of the proximal term (parameter μ) can be adjusted by the user, and it is important to choose its value correctly, especially when the data is highly heterogeneous. Compared to the original algorithm, FedProx limits the influence of local updates to a certain range from the global model.

Scaffold [5] is designed to mitigate the challenges posed by data heterogeneity among clients in an FL setting. This method leverages control variates, a variance reduction technique, to counteract the client drift phenomenon often occurring in local updates during the FL process. Scaffold uses control variates (variance reduction) to correct for the 'client-drift' in its local updates. There are three main steps

in Scaffold: local updates to the client model, local updates to the client control variate and aggregating the updates. If control variate is equal to zero, then Scaffold performs almost the same way as FedAvg.

The FedNova algorithm [6] does not average the final local models, but updates normalized by the amount of local work. This reduces the impact of the number of local steps performed on the client side on the overall result. In addition clients report the amount of local work along with model updates. This is usually a single scalar value, which adds minimal communication costs. The server normalizes the updates.

FedDyn is designed for dynamic knowledge extraction, with a special emphasis on local models and personalization of learning [7]. The average strategy is replaced with focus distillation, the algorithm is based on a modified loss function.

FedDist [8] has the ability to identify dissimilarities between neurons of different clients and allows the architecture to be adapted to the specificity of the data. Besides the related algorithms solve the problem of weight divergence [9], [10] and allow optimizing the balance between various indicators such as classification accuracy and communication efficiency [11].

A fundamentally new approach to solving the problem of local data heterogeneity was proposed by the developers of MOON [12], based on contrastive learning at the model level, not the data level. The key feature of the algorithm is the use of a special loss function and comparison of model representations. The current local and global models are compared for alignment, as well as a comparison with the previous state of the local model to avoid bias. Weights are adjusted based on contrastive loss. Despite the fact that MOON is characterized by high accuracy, and works stably with varying degrees of data heterogeneity, there is a number of limitations. In particular, it is necessary to store three models simultaneously, namely the current local, the previous local and global. In addition, extra calculations are required to calculate the contrastive loss.

There are also more complex aggregation algorithms, such as FedGA [13], which uses genetic algorithms to optimize weights. Aggregation occurs at the level of the base layers of models, and Gradient Alignment is used to solve the problems of error asymmetry. The algorithm shows high efficiency in heterogeneity conditions, achieves high accuracy and provides better model convergence compared to algorithms such as FedAvg and FedProx. However, the implementation of the algorithm increases the requirements for server resources, and genetic algorithms imply high computational complexity and the need to store population data.

In recent years, intelligent aggregation algorithms have been actively developed. For example, the IOFL algorithm [14] is proposed to improve the level of privacy protection and improve the performance of the global model when working with non-identically distributed data. An intelligent algorithm is used as an optimization method that searches for model parameters and distributes them among clients. Clients use local data to test the model received from the

server and send the testing results back to the server. The accuracy of testing on the client side serves as the basis for constructing a fitness function, which is a weighted average of the obtained results and is used to guide the optimization process.

The authors of the FedAAW algorithm [15] propose to solve the problem of data heterogeneity using analytical derivation of aggregation weights and adaptive adjustment of weights based on data quality. In this case, the upper convergence limit is minimized. The algorithm is designed to work in edge networks, cloud infrastructures and Internet of Things (IoT) systems.

The MOHFL algorithm [16] is based on the use of quality metrics for evaluating local models, a dynamic task distribution function between clients, and a load balancing mechanism. Therefore in this way, it is proposed to improve adaptability to various conditions and resilience to failures, optimize resource use, and improve training quality indicators.

SelfFed is based on the principles of self-control and self-learning to improve the efficiency and stability of models [17]. The algorithm assumes independent determination of optimal training parameters, local adaptation to the specifics of the data, automatic correction upon detection of errors, self-controlled aggregation of updates. For this purpose, mechanisms for automatic adjustment of hyper-parameters, adaptive error correction functions, self-monitoring metrics for training quality, and self-checking systems for data integrity are used.

FedAgent is based on the concept of intelligent agents to optimize the distributed learning process [18]. The algorithm operates by initializing agents on each device and distributing roles between them, coordinating actions and exchanging information between agents.

The conducted analysis shows that the problem of correct aggregation of local models is relevant for many researchers in the field of federated learning. It should be noted that most of the presented models are evaluated using such datasets as MNIST, CIFAR-10 and CIFAR-100 [19]. MNIST dataset is a set of images of handwritten digits written in different handwritings and with different degrees of accuracy. CIFAR-10 and CIFAR-100 data sets are sets of images of various objects.

Using the same data sets makes it easier to compare the proposed solutions, but the data sets themselves are only somewhat heterogeneous. Since modern distributed systems such as the IoT consist of different types of devices, special attention must be paid to data of different formats within a single system.

III. CHALLENGES

Currently, there are a number of challenges associated with non-identically distributed data in federated learning systems. In particular, there is no consensus on the classification and evaluation of non-identically distributed data, which makes it difficult to compare different studies and solutions. In addition, there are no common criteria or scenarios (e.g., on data partitioning and modeling) that

allow adequately testing new approaches to solving non-identically distributed data problems.

Another problem related to the first one is the lack of metrics for assessing heterogeneity. Surveys show that only a few researchers have used metrics to measure the level of non-identically distributed data. Assessing the degree of non-identically distributed data is important for proper data analysis in federated learning. Thus without such metrics, it is difficult to determine how serious the non-identically distributed data problems are and how effective the proposed solutions could be in practice.

Another issue is the interdependence of data shift types. When partitioning data to create a specific shift type (e.g., label shift), side effects in other shift types are often unintentional. However, note that researchers rarely consider these effects in their experimental studies. For example, in the reviewed papers, about half of the studies using label shift do not mention count shift, although the chosen partitioning methods cause such side effects. This highlights the complexity of non-identically distributed data scenarios and the need for a more comprehensive approach to data preparation and analysis.

It should also be noted that most of the works cover data characterized by label shifts. Although this type of shift has a significant impact on the quality of federated learning models, it is necessary to pay attention to other types of shifts as well. In addition, assessing only one type of shift is insufficient to fully understand and solve non-identically distributed data problems in FL. In real-world conditions, data is usually subject to several types of shifts simultaneously, and modeling such situations is important for developing robust solutions. Therefore, a more comprehensive approach to analyzing data non-identity is needed.

Finally, most of the studies and approaches considered are evaluated mainly on data that present image sets. Only in a few cases can we see an example of the algorithm working on other data set, for example, MovieLens-100K which consists of movie ratings produced by users [12] or network data [20]. Therefore, for modern heterogeneous systems, it is important to ensure correct work on aggregation of heterogeneous data.

IV. CONCLUSION

This paper presents the study of difficulties arising due to non-uniform distribution of data between participants in the federated learning processes. An important aspect of scientific novelty is a comprehensive consideration of the potential of aggregation methods in relation to situations with data non-uniformity.

As directions of further research on this topic, it is planned to develop and improve strategies for adapting the weighting coefficients of aggregation algorithms, including reducing the factors of differences between clients. In addition, it is additionally supposed to take into consideration the possibility of dynamic changes in hyper-parameters, including the learning speed on clients, data volumes and other characteristics, which is likely to reduce the negative effects of non-identical data distribution. It is also supposed to study and optimize the mechanisms of client clustering within federated learning based on the features of similarity of their data with individual learning

within each cluster. In addition, it is planned to pay additional attention to the study of issues of taking into account different levels of trust and levels of information security of clients in the general complex of issues of improving the quality of federated learning algorithms.

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